Experiment 'p_4_9_reverse_string' Results

June 3, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` : 0
Responses containing `expect` : 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_4_9_reverse_string
Natural language statement: Write a method that reads a word and returns the word in reverse.
Method signature: p_4_9_reverse_string(s: string) returns (reversed: string)

Ensures

- `|reversed| == |s|`
- `forall i :: 0 <= i < |s| ==> reversed[i] == s[|s| - 1 - i]`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method that reads a word and returns the word in reverse.

The signature should be:

p_4_9_reverse_string(s: string) returns (reversed: string)

The method should respect the following contract:

ensures |reversed| == |s|, ensures forall i :: 0 <= i < |s| ==> reversed[i] == s[|s| - 1
    ↪ - i]

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-Dmfemd5ur1otB0QdwOSGV0lfnRWhl
Model: gpt-5.5-2026-04-23
Created at: 1780492244
Finish reason: stop
Usage: 318 tokens in, and 6111 tokens out

```
{
  reversed := "";
  var i := 0;
  while i < |s|
    invariant 0 <= i <= |s|
    invariant |reversed| == i
    invariant forall j :: 0 <= j < i ==> reversed[j] == s[|s| - 1 - j]
  {
    ghost var oldI := i;
    ghost var oldReversed := reversed;
    assert oldI < |s|;
    assert |oldReversed| == oldI;
    assert forall j :: 0 <= j < oldI ==> oldReversed[j] == s[|s| - 1 - j];

    reversed := reversed + [s[|s| - 1 - i]];
    assert reversed == oldReversed + [s[|s| - 1 - oldI]];
    i := i + 1;
    assert i == oldI + 1;
    assert |reversed| == i;
```

```
    forall j | 0 <= j < i
      ensures reversed[j] == s[|s| - 1 - j]
    {
      if j < oldI {
        assert oldReversed[j] == s[|s| - 1 - j];
        assert reversed[j] == oldReversed[j];
      } else {
        assert j == oldI;
        assert reversed[j] == s[|s| - 1 - oldI];
        assert |s| - 1 - oldI == |s| - 1 - j;
      }
    }
  }
  assert i == |s|;
  assert forall j :: 0 <= j < |s| ==> reversed[j] == s[|s| - 1 - j];
}
```

Final Program

```
method p_4_9_reverse_string(s: string) returns (reversed: string)
       ensures |reversed| == |s|
       ensures forall i :: 0 <= i < |s| ==> reversed[i] == s[|s| - 1 - i]
{
  reversed := "";
  var i := 0;
  while i < |s|
    invariant 0 <= i <= |s|
    invariant |reversed| == i
    invariant forall j :: 0 <= j < i ==> reversed[j] == s[|s| - 1 - j]
  {
    ghost var oldI := i;
    ghost var oldReversed := reversed;
    assert oldI < |s|;
    assert |oldReversed| == oldI;
    assert forall j :: 0 <= j < oldI ==> oldReversed[j] == s[|s| - 1 - j];

    reversed := reversed + [s[|s| - 1 - i]];
    assert reversed == oldReversed + [s[|s| - 1 - oldI]];
    i := i + 1;
    assert i == oldI + 1;
    assert |reversed| == i;

    forall j | 0 <= j < i
      ensures reversed[j] == s[|s| - 1 - j]
    {
      if j < oldI {
        assert oldReversed[j] == s[|s| - 1 - j];
        assert reversed[j] == oldReversed[j];
      } else {
        assert j == oldI;
```

```
      assert reversed[j] == s[|s| - 1 - oldI];
      assert |s| - 1 - oldI == |s| - 1 - j;
    }
   }
  }
  assert i == |s|;
  assert forall j :: 0 <= j < |s| ==> reversed[j] == s[|s| - 1 - j];
}
```

Total Token Usage

Input tokens: 318
Output tokens: 6111
Reasoning tokens: 5631
Sum of 'total tokens': 6429

Experiment Timings

Overall Experiment started at 1780492244669, ended at 1780492356131, lasting 111462ms (111.46 seconds)
Iteration #1 started at 1780492244672, ended at 1780492356131, lasting 111459ms (111.46 seconds)